ECO5050 AI Applications and Techniques in Economics

Lecture 4 Basic Time Series Concepts and Linear Prediction Models

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Fall 2025

Outline

Basic Time-Series Concepts

- There are two forms of stationarity for a stochastic process (or time series)
  1. Strictly stationary
  2. Covariance (or second-order) stationary

Strictly stationary stochastic process

Consider the stochastic process $\{X_t\}_{t=1}^{\infty}$, where $f(x_{t_1}, x_{t_2}, \ldots, x_{t_n})$ is the joint density of the sequence $\{X_{t_1}, X_{t_2}, \ldots, X_{t_n}\}$, and $f(x_{t_1+j}, x_{t_2+j}, \ldots, x_{t_n+j})$ is the joint density of the sequence $\{X_{t_1+j}, X_{t_2+j}, \ldots, X_{t_n+j}\}$. The stochastic process $\{X_t\}$ is **strictly stationary** if

$$f(x_{t_1}, x_{t_2}, \ldots, x_{t_n}) = f(x_{t_1+j}, x_{t_2+j}, \ldots, x_{t_n+j})$$

all integers $j \geq 1$.

- Let $t_1 = 1$ and $n = 1$, then X_t has the same distribution as X_1 for all $t = 2, 3, \ldots \Rightarrow$ Strict stationarity implies that the sequence $\{X_t\}_{t=1}^{\infty}$ is identically distributed.

Basic Time-Series Concepts

Covariance stationary stochastic process

A stochastic process $\{X_t\}_{t=1}^{\infty}$ with a finite second moment $E(X_t^2) < \infty$ is **covariance stationary** if

- $E(X_t) = \mu$ is constant
- $\text{Cov}(X_t, X_{t+j}) = \gamma_j$ depends only on j and not t

- Strictly stationary processes that has a finite second moment is covariance stationary.
- It is possible that a covariance stationary process is not strictly stationary; the mean and covariance could not be functions of time, but perhaps higher moments are.

Basic Time-Series Concepts

- Covariance stationary processes can be built from white noise.
- A **white noise** process is a sequence $\{\varepsilon_t\}_{t=-\infty}^{\infty}$ that satisfies

$$
\begin{gathered}
E(\varepsilon_t) = 0 \\
E(\varepsilon_t^2) = \sigma^2 \\
E(\varepsilon_t \varepsilon_s) = 0 \text{ for all } t \neq s.
\end{gathered}
$$

Denote $\varepsilon_t \sim WN(0, \sigma^2)$ for a white noise.

Basic Time-Series Concepts

- The lag operator L satisfies $LY_t = Y_{t-1}$
- The lag operator can be applied multiple times to a series:

$$L^k Y_t = LL \cdots LY_t = Y_{t-k}$$

- An operator defined as a polynomial in L is called a lag polynomial

$$\begin{aligned} Y_t &= \mu + \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \cdots \\ &= \mu + \psi_0 \varepsilon_t + \psi_1 L \varepsilon_t + \psi_2 L^2 \varepsilon_t + \cdots \\ &= \mu + (\psi_0 + \psi_1 L + \psi_2 L^2 + \cdots) \varepsilon_t \\ &= \mu + \psi(L) \varepsilon_t \end{aligned}$$

Linear Projection

- We know that under squared error loss, the best predictor of Y is the conditional expectation $E[Y_{t+1}|Z_t]$.
- It requires the knowledge of the joint distribution of (Y_{t+1}, Z_t) to calculate $E[Y_{t+1}|Z_t]$.
- Restrict the predictor of Y as a linear function of Z; we define an approximation to the conditional expectation function by the linear function.
- The **linear projection** of Y_{t+1} on Z_t is the forecast that produces the smallest MSE among the class of linear forecast models:

$$\hat{E}[Y_{t+1}|Z_t] = Z_t'\beta^*$$

  where β^* satisfies the orthogonality condition

$$E[Z_t(Y_{t+1} - Z_t'\beta^*)] = \mathbf{0}$$

- The linear projection $\hat{E}[Y_{t+1}|Z_t]$ is the optimal linear forecast.

Linear Projection

- Assume $\{Y_t\}$ is covariance stationary and let $Z_t = (Y_t, Y_{t-1}, Y_{t-2}, \ldots)$ be the infinite past history.
- The projection error

$$\varepsilon_t = Y_t - \hat{E}[Y_t | Y_{t-1}, Y_{t-2}, \ldots]$$

  is a white noise process. It satisfies

$$E(\varepsilon_t) = 0$$
$$E(\varepsilon_t \varepsilon_{t-j}) = 0$$
$$\sigma^2 = E(\varepsilon_t^2) \leq E(Y_t^2) < \infty$$

- A covariance stationary process $\{Y_t\}$ has a white noise projection error. The Wold representation theorem states that Y_t can be expressed as a infinite linear function of the projection errors.

Wold representation theorem

Wold representation theorem

Any covariance stationary stochastic process $\{Y_t\}$ can be represented in the form as a linear combination of white noise terms ε_{t-j}, $j = 0, \ldots, \infty$ and a deterministic component μ_t:

$$Y_t = \mu_t + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$$

where ε_t are the white noise projection errors

$$\varepsilon_t \equiv Y_t - \hat{E}[Y_t | Y_{t-1}, Y_{t-2}, \ldots].$$

and $\{\psi_j\}$ are independent of time, $\theta_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

- In most cases $\mu_t = \mu$, a constant. If so, Y_t is nondeterministic.

Wold representation theorem

- The Wold representation/decomposition theorem states that any covariance stationary process can be represented by the MA(∞) processes, which justifies linear models as approximations.
- It describes only optimal linear forecasts of Y_t, it may be that a nonlinear model provides a better approximation.
- A practical concern is that the moving average order is potentially infinite.
- Under the assumption that the coefficients ψ_j in $\psi(L)$ is absolutely summable so that $\sum_{j=0}^{\infty} |\psi_j| < \infty$, it is possible to approximate $\psi(L)$ by a ratio of two finite-order polynomials $\theta(L)/\phi(L)$. This yields the general form of an ARMA(p, q) process.
- Hence a popular modeling strategy is to approximate the MA representation implied by the Wold representation with an ARMA(p, q) model where both p and q are of low order.

ARMA Process

- $\{Y_t\}$ that follows the following model is called an ARMA(p, q) process.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$
$$\phi(L) Y_t = \theta(L) \varepsilon_t$$

- The MA part of the process, $\sum_{i=0}^{q} \theta_i \varepsilon_{t-i}$ is stationary for any finite q. Hence Y_t is stationary if the AR part of the process is stationary.

Autoregressive Process

- $\{Y_t\}$ that follows a pth-order autoregression, denoted AR(p), satisfies

$$Y_t = \mu + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t$$
$$\phi(L) Y_t = \mu + \varepsilon_t$$

  where $\varepsilon_t \sim WN(0, \sigma^2)$ and $\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p$ is a lag polynomial.

- The AR(p) process is stationary only if L^* such that $\phi(L^*) = 0$ satisfies $|L^*| > 1$.

- Example: The AR(1) model $Y_t = \phi_1 Y_{t-1} + \varepsilon_t$ is stationary if $|\phi_1| < 1$.

Moving Average Process

- A MA(q) process can be written as

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$
$$Y_t = \mu + \theta(L)\varepsilon_t$$

  where $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$.

- As long as q is finite, MA(q) processes are always stationary.
- If a MA(q) process can be written as an AR(∞) representation, the MA process is **invertible**.
- The MA(q) process is invertible only if L^* such that $\theta(L^*) = 0$ satisfies $|L^*| > 1$.
- Example: The MA(1) model $Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$ is invertible if $|\theta_1| < 1$

ARMA Process

- The ARMA(p, q) process.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$
$$\phi(L) Y_t = \theta(L) \varepsilon_t$$

- The stationarity and invertibility conditions of an ARMA(p, q):
  - Stationary: $\phi(L^*) = 0 \Longrightarrow |L^*| > 1$
  - Invertible: $\theta(L^*) = 0 \Longrightarrow |L^*| > 1$

- An ARMA(p, q) process that is stationary and invertible can be written either as an AR model or as an MA model, typically of infinite order.

$$Y_t = \phi(L)^{-1} \theta(L) \varepsilon_t$$
$$\theta(L)^{-1} \phi(L) Y_t = \varepsilon_t$$

Estimation and Lag Selection for ARMA models

- ARMA models are based on linear projections which provide reasonable forecasts of linear processes under MSE loss. ARMA forecasts require
  - the order of the ARMA(p, q) model
  - estimates of the parameters $\phi_1, \ldots, \phi_q, \theta_1, \ldots, \theta_q$
  - estimates of past "shocks" $\varepsilon_t, \ldots, \varepsilon_{t-q}$ if $q \geq 1$
- The common approach to estimate the parameters is to use maximum likelihood estimation (MLE) under the assumption of normal errors.
- A state-space form can be used for constructing MLE, which has a number of advantages.
  - When $q > 0$, the state-space form and the Kalman filter provides an easy estimation strategy
  - The state-space representation provides a convenient representation of the model for multistep forecasts
- The common approach in selecting the lag orders p and q is the model selection criteria, such as IC

Forecasting with ARMA models

- To introduce how the ARMA(p, q) forecasts are generated, lets treat the parameters as known (ignore estimation error)
- Consider $q = 0$, the AR(p) model:

$$y_{t+1} = c + \phi_1 y_t + \cdots + \phi_p y_{t-p+1} + \varepsilon_{t+1}, \;\; \varepsilon_{t+1} \sim WN(0, \sigma^2)$$

  where $Z_t = (y_t, \ldots, y_1)$.
- Since $\hat{E}[\varepsilon_{t+1}|Z_t] = 0$ and $\{y_t, \ldots, y_{t-p+1}\} \in Z_t$, the linear forecast of y_{t+1} given Z_t is

$$f_{t+1|t}(Z_t, \beta) = c + \phi_1 y_t + \cdots + \phi_p y_{t-p+1}$$

  where $\beta = (c, \phi_1, \ldots, \phi_p)$.

Forecasting with ARMA models

- Multiperiod forecasts are generated by replacing unknown values of y_{t+h} $(h \geq 1)$ with their forecasts $f_{t+h|t}(Z_t, \beta)$, setting up a recursive system:

$$
\begin{aligned}
f_{t+2|t} &= c + \phi_1 f_{t+1|t} + \phi_2 y_t + \cdots + \phi_p y_{t-p+2}, \\
f_{t+3|t} &= c + \phi_1 f_{t+2|t} + \phi_2 f_{t+1|t} + \phi_3 y_t + \cdots + \phi_p y_{t-p+3}, \\
&\vdots \\
f_{t+p+1|t} &= c + \phi_1 f_{t+p|t} + \phi_2 f_{t+p-1|t} + \phi_3 f_{t+p-2|t} + \cdots + \phi_p f_{t+1|t}
\end{aligned}
$$

- Example: Forecasts generated by the AR(1) model $y_{t+1} = c + \phi_1 y_t + \varepsilon_{t+1}$

Forecasting with ARMA models: MA

- Consider the MA(q) model:

$$y_{t+1} = \mu + \varepsilon_{t+1} + \theta_1 \varepsilon_t + \cdots + \theta_q \varepsilon_{t-q+1}$$

- Assuming that $\mu, \theta_1, \ldots, \theta_q$ are known and $Z_t = (\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_0)$, the forecasts are

$$\begin{aligned}
f_{t+1|t} &= \mu + \theta_1 \varepsilon_t + \cdots + \theta_q \varepsilon_{t-q+1} \\
f_{t+2|t} &= \mu + \theta_2 \varepsilon_t + \cdots + \theta_q \varepsilon_{t-q+2}, \\
&\vdots \\
f_{t+q|t} &= \mu + \theta_q \varepsilon_t \\
f_{t+q+1|t} &= \mu
\end{aligned}$$

- The MA(q) has limited memory; values more than q periods into the future is simply the unconditional mean of the process $E(Y_t) = \mu$.

Forecasting with ARMA models: MA

- To compute forecasts we need to estimate $\varepsilon_t, \varepsilon_{t-1}, \ldots$ from the realizations $y_t, y_{t-1}, \ldots$. This can be achieved assuming the invertibility condition.

$$\varepsilon_t = \theta(L)^{-1} y_t = \alpha(L) y_t$$

where $\theta(L) = \theta_0 + \theta_1 L + \cdots + \theta_q L^q$, $\theta_0 = 1$ and $\theta(L)\alpha(L) = 1$. For simplicity assume $\mu = 0$.

- The coefficients of $\alpha(L) = \alpha_0 + \alpha_1 L + \alpha_2 L^2 + \cdots$ can be obtained recursively using the relations

$$\begin{aligned}
\alpha_0\theta_0 &= 1 \\
\alpha_1\theta_0 + \alpha_1\theta_1 &= 0 \\
&\vdots \\
\alpha_q\theta_0 + \alpha_{q-1}\theta_1 + \cdots + \alpha_0\theta_q &= 0 \\
\alpha_i\theta_0 + \alpha_{i-1}\theta_1 + \cdots + \alpha_0\theta_i &= 0 \text{ for } i > q
\end{aligned}$$

- The unobserved shocks can be written entirely as functions of the parameter values and current and past values of y.

Forecasting with ARMA models: MA

- Example: Forecasts generated by the MA(1) model $y_{t+1} = \varepsilon_{t+1} + \theta_1 \varepsilon_t$. Under invertiblity $\varepsilon_{t+1} = \alpha(L) y_{t+1}$,

$$(1 + \theta_1 L)(\alpha_0 + \alpha_1 L + \alpha_2 L^2 + \cdots) = 1$$
$$\alpha(L) = 1 - \theta_1 L + \theta_1^2 L^2 - \theta_1^3 L^3 + \cdots$$

- The forecasts are

$$f_{t+1|t} = \theta_1 \varepsilon_t = \theta_1 (1 - \theta_1 L + \theta_1^2 L^2 - \theta_1^3 L^3 + \cdots) y_t$$
$$f_{t+h|t} = 0 \text{ for } h > 1$$

- Recursions using past forecasts can also be employed to update the forecasts:
  - Since $f_{t|t-1} = \hat{E}[y_t | z_{t-1}] = \theta_1 \varepsilon_{t-1}$, we have $\varepsilon_t = y_t - f_{t|t-1}$
  - $f_{t+1|t} = \theta_1 \varepsilon_t = \theta_1 (y_t - f_{t|t-1})$

Forecasting with ARMA models

- This procedure can be adapted to forecasting using ARMA models. $\phi(L)y_t = \theta(L)\varepsilon_t$

$$\varepsilon_t = \theta(L)^{-1}\phi(L)y_t = \alpha(L)y_t$$

$$\begin{aligned}
\phi(L)f_{t+1|t} &= \theta_1\alpha(L)y_t + \theta_2\alpha(L)y_{t-1} + \cdots + \theta_q\alpha(L)y_{t-q+1} \\
\phi(L)f_{t+2|t} &= \theta_2\alpha(L)y_t + \theta_3\alpha(L)y_{t-1} + \cdots + \theta_q\alpha(L)y_{t-q+2} \\
&\vdots \\
\phi(L)f_{t+h|t} &= 0 \text{ for } h > q
\end{aligned}$$

Direct vs. Iterated Multiperiod Forecasts

- In obtaining multiperiod forecasts in the context of AR models, there is a question of whether an iterated or a direct forecasting approach should be used.
- Suppose the observations $y_1, \ldots, y_T$ follow the stationary AR(1) model

$$y_t = c + \phi y_{t-1} + \varepsilon_t, \quad |\phi| < 1, \varepsilon_t \overset{i.i.d.}{\sim} (0, \sigma^2)$$

- The stationary AR(1) model can be be expressed as

$$\begin{aligned} y_t &= c\left(\frac{1-\phi^h}{1-\phi}\right) + \phi^h y_{t-h} + \sum_{j=0}^{h-1} \phi^j \varepsilon_{t-j} \\ &= c_h + \phi_h y_{t-h} + v_t \end{aligned}$$

The error term v_t follows an MA$(h-1)$ process even when ε_t is serially uncorrelated due to the data overlap resulting from $h > 1$.

Direct vs. Iterated Multiperiod Forecasts

- The direct approach is to estimate the parameters c_h and ϕ_h directly by regressing y_t on y_{t-h}, and use them to construct the forecast

$$f_{T+h|T} = \hat{c}_h + \hat{\phi}_h y_T$$

- The iterated approach plugs in the parameter estimates of the single-period model $\hat{c}$ and $\hat{\phi}$ and iterate forward to the desired horizon using chain rules

$$f_{T+h|T} = \hat{c}\left(\frac{1-\hat{\phi}^h}{1-\hat{\phi}}\right) + \hat{\phi}^h y_T$$

Direct vs. Iterated Multiperiod Forecasts

- In comparing the predictive accuracy of iterated and direct forecasting methods, theoretical analysis suggests a bias and variance trade-off.
  - When the AR model is correctly specified, the iterated approach makes more efficient use of the data and will have smaller MSE than the direct method.
  - Under misspecification, the direct approach tends to be more robust; the misspecification error from the iterated forecasts are compounded in longer horizons and lead to larger bias.
- Which approach performs best will depend on the true DGP, the degree of model misspecification, both of which are unknown, as well as the parameter estimation error which reflects the sample size.
  - Marcellino, Stock, and Watson (2006) suggests empirical evidence that the iterated approach outperform direct forecasts, particularly if the models can select longer lag length.
  - Pesaran, Pick, and Timmermann (2011)

Vector Autoregressions

- Vector Autoregressions (VAR) generalize the univariate autoregressions to the multivariate case where y_t becomes a $(k \times 1)$ vector and the information set is extended to $Z_t = \{y_{i,t}, y_{i,t-1}, \ldots\}_{i=1}^{k}$.
- As with univariate data, nearly all work on forecasting with VARs assumes squared error loss and uses VARs to approximate the conditional mean of the outcome variable of interest.
- The Wold representation theorem extends to the multivariate case and hence VARs can be used to approximate covariance stationary multivariate processes.

Vector Autoregressions

- The pth order VAR for a $(k \times 1)$ vector $\mathbf{y}_t$:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \cdots + \mathbf{\Phi}_p \mathbf{y}_{t-p} + \mathbf{u}_t$$

  where $\mathbf{\Phi}_i$ is a $(k \times k)$ matrix of autoregressive coefficients for $i = 1, \ldots, p$, and $E[\mathbf{u}_t \mathbf{u}_t'] = \Sigma$.

- The elements of $\mathbf{u}_t$, $u_{i,t}$ can be correlated across i but the lag length p is typically chosen to be large enough so $\mathbf{u}_t$ is serially uncorrelated.

- VAR is a system of equations with the same regressors appearing in each equation

$$y_{i,t} = c_i + \sum_{j=1}^{p} \phi_{1ij} y_{1,t-j} + \cdots + \sum_{j=1}^{p} \phi_{kij} y_{k,t-j} + u_{i,t}, \quad i = 1, \ldots, k$$

  here the first p terms show how $y_{i,t}$ is affected by lags of y_1, while the last p terms show how it is affected by lags of y_k.

Vector Autoregressions

- VARs became a popular forecasting method because of their relative simplicity in terms of which choices need to be made by the forecaster: which variables to include, lag order p.
- However the risk of overparameterization of VARs is high, the VAR(p) has $k^2p + k$ coefficients and $k(k+1)/2$ covariance parameters in Σ to estimate.
- We can use methods such as Granger causality tests to remove variables that do not appear to be useful and hence reduce the size of the model.
- The standard approach for choosing the lag order p is to treat this as a model selection problem and use information criteria such as BIC and AIC.

Vector Autoregressions

Estimation of VAR(p)

- VARs are seemingly unrelated regression (SUR) equations whose equations are related through the covariance Σ.
- When each equation has the same regressors such as the VAR, SUR estimation simplifies to be numerically identical to OLS estimation equation by equation.
- Let $\mathbf{X}_t \equiv (\mathbf{Y}'_{t-1}, \mathbf{Y}'_{t-1}, \ldots, \mathbf{Y}'_{t-p})$, $\Pi \equiv (\mathbf{c}\ \mathbf{\Phi}_1 \cdots \mathbf{\Phi}_p)$, then the VAR($p$) model can be written as

$$\mathbf{Y}_t = \mathbf{\Pi}\mathbf{X}_t + \mathbf{u}_t$$

  where sizes are $(k \times 1)$, $(k \times (kp+1))$, $((kp+1) \times 1)$, and $(k \times 1)$ respectively.
- Under Gaussian errors $\mathbf{u}_t \overset{i.i.d.}{\sim} N(0, \Sigma)$, the MLE of $\mathbf{\Pi}$ can be computed by OLS regressions of $\mathbf{Y}_t$ on $\mathbf{X}_t$

Vector Autoregressions

Estimation of VAR(p)

- The log-likelihood is

$$\log L = -\frac{Tk}{2}\log(2\pi) - \frac{T}{2}\log|\Sigma| - \frac{1}{2}\sum_t (\mathbf{Y}_t - \mathbf{\Pi}\mathbf{X}_t)'\Sigma^{-1}(\mathbf{Y}_t - \mathbf{\Pi}\mathbf{X}_t)$$

- The MLE $\hat{\mathbf{\Pi}}$ is

$$\hat{\Pi} = \sum_t \mathbf{Y}_t\mathbf{X}_t' \left(\sum_t \mathbf{X}_t\mathbf{X}_t'\right)^{-1}$$

and

$$\hat{\Sigma} = \frac{1}{T}\sum_t \hat{\mathbf{u}}_t\hat{\mathbf{u}}_t' \quad \text{where } \hat{\mathbf{u}}_t = \mathbf{Y}_t - \hat{\mathbf{\Pi}}\mathbf{X}_t$$

Vector Autoregressions

Granger causality

- Consider the case of two variables: the question is whether a scalar y can help forecast another scalar x. If it cannot, we say that y does not Granger-cause x.

Granger causality

y fails to Granger-cause x if for all $h > 0$ the mean squared error (MSE) of a forecast of x_{t+h} based on $(x_t, x_{t-1}, \ldots)$ is the same as the MSE of a forecast of x_{t+h} that uses both $(x_t, x_{t-1}, \ldots)$ and $(y_t, y_{t-1}, \ldots)$. If we restrict ourselves to linear functions, y fails to Granger-cause x if

$$MSE[\hat{E}(x_{t+h}|x_t, x_{t-1}, \ldots)] = MSE[\hat{E}(x_{t+h}|x_t, x_{t-1}, \ldots, y_t, y_{t-1}, \ldots)]$$

- Granger's (1969) reason for proposing this definition was that if an event Y is the cause of another event X, then the event Y should precede the event X.

- It is important to consider all possible variables that interact to bring about an outcome before making any definite conclusions about Granger causality; it is entirely possible that a finding of Granger causality may be overturned by adding more variables.
- Consider the bivariate VAR

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \phi_{11}(L) & \phi_{12}(L) \\ \phi_{21}(L) & \phi_{22}(L) \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}$$

If y_{2t} fails to Granger-cause y_{1t}, then $\phi_{12}(L) = 0$

Vector Autoregressions

Choice of lag order

- Assuming a constant $\mathbf{c}$ is included in the VAR system, with k variables, p lags, and T observations, the AIC and BIC information criteria take the forms

$$\text{AIC}(p) = \log|\hat{\Sigma}_p| + k(kp+1)\frac{2}{T}$$

$$\text{BIC}(p) = \log|\hat{\Sigma}_p| + k(kp+1)\frac{\ln(T)}{T}$$

  where $\hat{\Sigma}_p = T^{-1}\sum_{t=1}^{T}\hat{\mathbf{u}}_t\hat{\mathbf{u}}_t'$.

- The objective is to identify the model (lag order p) that minimizes the IC.
- In practice the search is sometimes restricted to include a minimum value of p that reflects the periodicity of the data.
  - When the data are quarterly a minimum of four lags are chosen to reflect seasonality patterns believed to be present in the data.

Vector Autoregressions

Multiperiod forecasts

- The multiperiod forecast for the VAR(1) model

$$\mathbf{Y}_{t+1} = \mathbf{c} + \boldsymbol{\Phi}\mathbf{Y}_{t-1} + \mathbf{u}_{t+1}, \quad \mathbf{u}_{t+1} \sim (\mathbf{0}, \Sigma)$$

$$\mathbf{Y}_{t+h} = (\mathbf{I}_k - \boldsymbol{\Phi}^h)(\mathbf{I}_k - \boldsymbol{\Phi})^{-1}\mathbf{c} + \boldsymbol{\Phi}^h\mathbf{Y}_t + \sum_{i=1}^{h} \boldsymbol{\Phi}^{h-i}\mathbf{u}_{t+i}$$

so the forecast under MSE loss becomes

$$f_{t+h|t} = (\mathbf{I}_k - \boldsymbol{\Phi}^h)(\mathbf{I}_k - \boldsymbol{\Phi})^{-1}\mathbf{c} + \boldsymbol{\Phi}^h\mathbf{Y}_t$$

Vector Autoregressions

Multiperiod forecasts

- For higher order $p > 1$, we can write VAR models in the companion form and then iterate the model h steps ahead
- Example: VAR(2) in the companion form

$$\begin{pmatrix} \mathbf{Y}_{t+1} \\ \mathbf{Y}_t \end{pmatrix} = \begin{pmatrix} \mathbf{c} \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 \\ \mathbf{I}_k & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \end{pmatrix} \begin{pmatrix} \mathbf{u}_{t+1} \\ \mathbf{0} \end{pmatrix}$$

$$\tilde{\mathbf{Y}}_{t+1} = \tilde{\mathbf{c}} + \tilde{\mathbf{\Phi}} \tilde{\mathbf{Y}}_t + \tilde{\mathbf{u}}_{t+1}$$

- Or we can use a loss function defined directly on the h-step forecast error (direct approach) and estimate the following for the VAR(p) case

$$\mathbf{Y}_{t+h} = \mathbf{c} + \sum_{i=1}^{h} \mathbf{\Phi}_{hi} \mathbf{Y}_{t+1-i} + \mathbf{u}_{t+h}$$

Vector Autoregressions

- Factor-Augmented VARs (FAVAR)
  - If k is very large, prediction models using unrestricted VARs involve a very large number of parameters to estimate.
  - Bernanke, Boivin, and Eliasz (2005) proposed the FAVAR, which augment a small set of common factors $\mathbf{F}_t$ with key variables of interest $\mathbf{Y}_t$. Let $\tilde{\mathbf{Y}}_t = (\mathbf{Y}'_t, \mathbf{F}'_t)'$

$$\tilde{\mathbf{Y}}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{\Phi}_i \tilde{\mathbf{Y}}_{t-i} + \tilde{\mathbf{u}}_t$$

  - In practice $\mathbf{F}_t$ will be extracted from a set of factors, which we will learn more in detail.
- Bayesian VARs
  - Bayesian forecasts can be though of as shrinkage methods that reduce the effect of estimation errors.
  - It requires choice of priors and algorithms for estimation. Bayesian forecasts are not covered in this course.